08_Class_Activity

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In class activity 8: Study Design and Power Analysis

Introduction

This activity demonstrates key concepts in experimental design using a made up fish example:

1. **Formulating research questions**
2. **Understanding study designs**
3. **Recognizing proper replication vs. pseudoreplication**
4. **Conducting power analysis** (before and after data collection)
5. **Planning sampling strategies**

We'll study fish mass above and below waterfalls across multiple streams.

Part 1: Load Required Packages

```
library(tidyverse)
library(pwr)

set.seed(42)
```

Part 2: Research Question

Main Question: Do waterfalls affect fish mass in stream ecosystems?

Hypothesis: Fish below waterfalls are larger due to better feeding opportunities from nutrients and prey from the lake.

Part 3: Study Design - Natural Experiment

We'll sample fish above and below waterfalls in 6 different streams. This represents a "natural experiment" since we can't manipulate waterfall presence.

```
# Create fish data from streams
# ADJUST THESE VARIABLES TO EXPLORE DIFFERENT SCENARIOS:

n_streams <- 5        # Number of streams to sample
n_fish_per_site <- 6  # Number of fish per site (above/below)
waterfall_effect <- 15 # Effect size: how much larger fish are below (in grams)

# Stream characteristics (baseline means will be generated)
stream_df <- tibble(
  stream_id = 1:n_streams,
  stream_mean = rnorm(n_streams, mean = 85, sd = 5)  # Random baseline for each stream
)

f_df <- tibble(
  stream_id = rep(1:n_streams, each = n_fish_per_site * 2),
  position = rep(c("above", "below"), each = n_fish_per_site, times = n_streams),
```

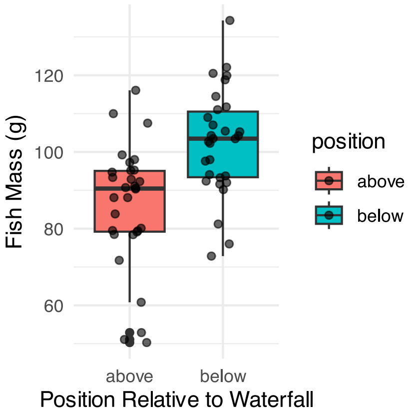
```
  fish_id = 1:(n_streams * n_fish_per_site * 2)
) %>%
  left_join(stream_df, by = "stream_id") %>%
  mutate(
    # Above waterfall = stream baseline, below = baseline + effect
    expected_mass = ifelse(position == "above", stream_mean, stream_mean +
waterfall_effect),
    mass_g = rnorm(n(), mean = expected_mass, sd = 12)
  ) %>%
  select(stream_id, position, fish_id, mass_g)

head(f_df, 12)
```

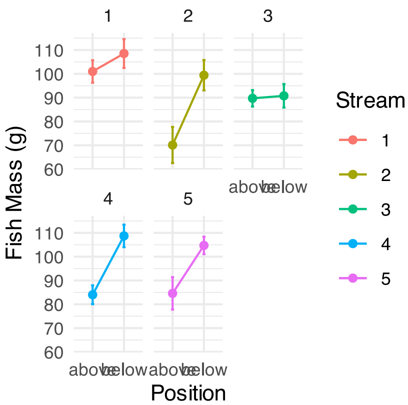
```
# A tibble: 12 × 4
   stream_id position fish_id mass_g
       <int> <chr>      <int>  <dbl>
 1         1 above          1   90.6
 2         1 above          2  110.
 3         1 above          3   90.7
 4         1 above          4  116.
 5         1 above          5   91.1
 6         1 above          6  108.
 7         1 below          7  134.
 8         1 below          8   90.2
 9         1 below          9  104.
10         1 below         10  105.
11         1 below         11  114.
12         1 below         12  103.
```

```
# Visualize the data
basic_plot <- f_df %>%
  ggplot(aes(x = position, y = mass_g, fill = position)) +
  geom_boxplot() +
  geom_jitter(width = 0.2, alpha = 0.6) +
  labs(x = "Position Relative to Waterfall",
       y = "Fish Mass (g)") +
  theme_minimal()

basic_plot
```



```r
# Show data by individual streams
stream_plot <- f_df %>%
  ggplot(aes(x = position, y = mass_g, group = stream_id, color = as.factor(stream_id))) +
  stat_summary(fun = mean, geom = "point") +
  stat_summary(fun = mean, geom = "line") +
 stat_summary(fun.data = mean_se, geom = "errorbar", width = 0.1) +
  facet_wrap(~stream_id) +
  labs(x = "Position", y = "Fish Mass (g)", color = "Stream") +
  theme_minimal()
stream_plot
```



Part 4: Replication Issues

What is the experimental unit here?

! Activity 1: Identify the Experimental Unit

Question: In our fish study, what is the true experimental unit?

A) Individual fish
B) Stream locations (above/below pairs)
C) Individual streams
D) All fish combined

Pseudoreplication Example

```r
# WRONG analysis: treating all fish as independent
# This ignores that fish within streams may be similar

pseudo_test <- t.test(mass_g ~ position, data = f_df)
pseudo_test
```

```

    Welch Two Sample t-test

data:  mass_g by position
t = -4.2726, df = 56.515, p-value = 7.488e-05
alternative hypothesis: true difference in means between group above and group below is not
equal to 0
95 percent confidence interval:
```

```
 -24.308784  -8.792322
sample estimates:
mean in group above mean in group below
           85.88446           102.43501
```

```
# CORRECT analysis: average by stream first, then compare
stream_means <- f_df %>%
  group_by(stream_id, position) %>%
  summarize(mean_mass = mean(mass_g), .groups = "drop")

# Reshape for paired analysis
stream_wide <- stream_means %>%
  pivot_wider(names_from = position,
              values_from = mean_mass)

proper_test <- t.test(stream_wide$below, stream_wide$above, paired = TRUE)
proper_test
```

```


    Paired t-test

data:  stream_wide$below and stream_wide$above
t = 3.1164, df = 4, p-value = 0.03565
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
  1.805299 31.295806
sample estimates:
mean difference
       16.55055
```

The proper analysis uses **streams as replicates** (n = 6), not individual fish (n = 36 per group).

Part 5: Power Analysis - Planning Phase

Let's calculate how many streams we need to detect a meaningful difference. The effect size here is called Cohen's D measures the difference between two group means in terms of standard deviations. It helps to understand the magnitude of a difference beyond statistical significance by expressing the distance between means as a standardized value. For example, a Cohen's d of 0.2 is considered a small effect, 0.5 a moderate effect, and 0.8 a large effect

```
# Expected values based on pilot data
above_mean <- 85
below_mean <- 110
pooled_sd <- 20

# Calculate effect size
effect_val <- abs(below_mean - above_mean) / pooled_sd
effect_val
```

```
[1] 1.25
```

```r
# Calculate required sample size for 80% power
power_result <- pwr.t.test(
  d = effect_val,
  sig.level = 0.05,
  power = 0.8,
  type = "paired"
)

power_result
```

```

     Paired t test power calculation 

              n = 7.171657
              d = 1.25
      sig.level = 0.05
          power = 0.8
    alternative = two.sided

NOTE: n is number of *pairs*
```

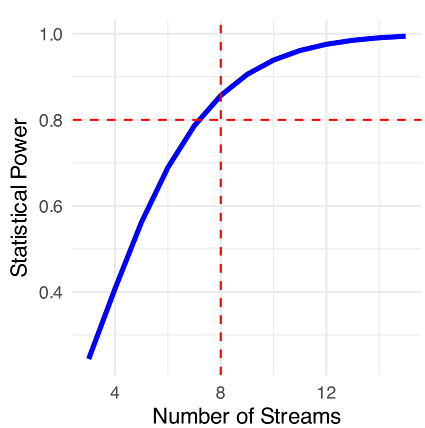
We need at least 7 streams for 80% power.

Power Curve

```r
# Show how power changes with sample size
power_curve_df <- tibble(streams = 3:15) %>%
  rowwise() %>%
  mutate(power = pwr.t.test(n = streams,
                            d = effect_val,
                            sig.level = 0.05,
                            type = "paired")$power) %>%
  ungroup()

curve_plot <- ggplot(power_curve_df, aes(x = streams, y = power)) +
  geom_line(linewidth = 1.2, color = "blue") +
  geom_hline(yintercept = 0.8, linetype = "dashed", color = "red") +
  geom_vline(xintercept = ceiling(power_result$n), linetype = "dashed", color = "red") +
  labs(x = "Number of Streams", y = "Statistical Power") +
  theme_minimal()

curve_plot
```



! Activity 2: Power Exploration

Change the values below to see how power changes:

```
# Experiment with different scenarios
above_mean <- 85
below_mean <- 100    # Try changing this
pooled_sd <- 25     # Try changing this

effect_val <- abs(below_mean - above_mean) / pooled_sd

power_result <- pwr.t.test(
  d = effect_val,
  sig.level = 0.05,
  power = 0.8,
  type = "paired"
)

power_result
```

Questions:

1. What happens to required sample size if the effect is smaller (below_mean = 95)?
2. What if variation is higher (pooled_sd = 30)?
3. What if we want 90% power instead of 80%?

Part 6: Post-Hoc Power Analysis

Now let's analyze the power we actually had with our 6 streams.

```
# Calculate observed effect size from our data
observed_above <- mean(stream_wide$above)
observed_below <- mean(stream_wide$below)
observed_diff <- observed_below - observed_above
observed_sd <- sd(stream_wide$below - stream_wide$above)

observed_effect <- observed_diff / observed_sd
observed_effect
```

```
[1] 1.393684
```

```
# What power did we actually have?
actual_power <- pwr.t.test(
  n = 6,
  d = observed_effect,
  sig.level = 0.05,
  type = "paired"
)

actual_power
```

```

     Paired t test power calculation
```

```
             n = 6
             d = 1.393684
     sig.level = 0.05
         power = 0.7778318
   alternative = two.sided

NOTE: n is number of *pairs*
```

With 6 streams, we had `100%` power to detect the effect we observed.

Part 7: Alternative Analysis Approaches

Unpaired t-test (ignoring pairing)

```r
# What if we ignored the stream pairing?
unpaired_test <- t.test(mean_mass ~ position, data = stream_means)
unpaired_test
```

```

	Welch Two Sample t-test

data:  mean_mass by position
t = -2.7496, df = 7.0247, p-value = 0.02842
alternative hypothesis: true difference in means between group above and group below is not
equal to 0
95 percent confidence interval:
 -30.773874  -2.327232
sample estimates:
mean in group above mean in group below 
           85.88446           102.43501 
```

Summary Statistics

```r
# Calculate summary by position
summary_df <- f_df %>%
  group_by(position) %>%
  summarize(
    n_fish = n(),
    mean_mass = mean(mass_g),
    sd_mass = sd(mass_g),
    se_mass = sd_mass / sqrt(n_fish)
  )

summary_df
```

```
# A tibble: 2 × 5
  position n_fish mean_mass sd_mass se_mass
  <chr>     <int>     <dbl>   <dbl>   <dbl>
1 above        30      85.9    16.2    2.95
2 below        30     102.     13.7    2.51
```

```r
# Summary by stream (proper replication level)
stream_summary <- stream_means %>%
  group_by(position) %>%
  summarize(
    n_streams = n(),
    mean_mass = mean(mean_mass),
    sd_mass = sd(mean_mass),
    se_mass = sd_mass / sqrt(n_streams)
  )

stream_summary
```

```
# A tibble: 2 × 5
  position n_streams mean_mass sd_mass se_mass
  <chr>        <int>     <dbl>   <dbl>   <dbl>
1 above            5      85.9      NA      NA
2 below            5     102.       NA      NA
```

Part 8: Design Your Own Study

! Activity 3: Study Design Practice

Scenario: You want to study if fish size differs between fast and slow water areas.

Design Questions:

1. **What is your experimental unit?** ____

2. **How many replicates do you need? Use these values:**

   - Fast water mean: 75g
   - Slow water mean: 85g
   - Standard deviation: 15g
   - Desired power: 80%

```r
# Calculate your required sample size
fast_mean <- 75
slow_mean <- 85
sd_val <- 15

effect_size <- abs(slow_mean - fast_mean) / sd_val

sample_result <- pwr.t.test(
  d = effect_size,
  sig.level = 0.05,
  power = 0.8,
  type = "two.sample"  # or "paired" if appropriate
)

sample_result
```

3. **What could cause pseudoreplication in this design?** ____

4. **How would you avoid it?** ____

Summary

Key Points:

- **Experimental unit**: The thing that receives the treatment independently
- **Replication**: Must be at the level of the experimental unit
- **Power analysis**: Plan sample size before collecting data
- **Paired vs unpaired**: Paired tests are more powerful when appropriate
- **Effect size**: Larger effects need fewer samples to detect

Common Mistakes:

- Treating subsamples as independent replicates
- Analyzing data at the wrong level
- Collecting data before planning analysis
- Ignoring natural pairing in the design